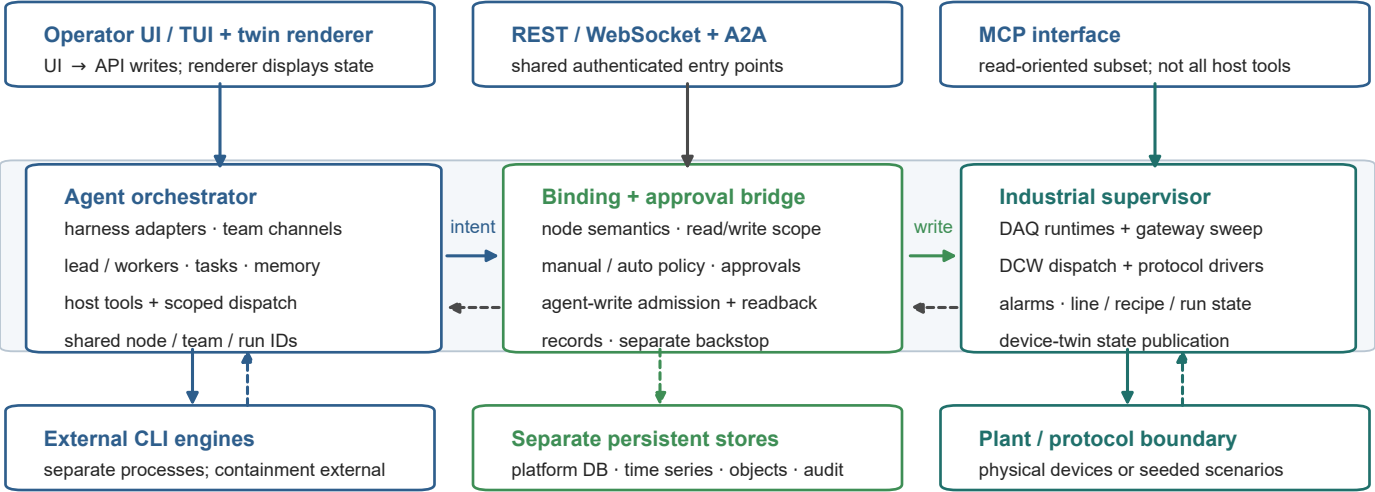
Operator UI / TUI + twin renderer
UI → API writes; renderer displays state
REST / WebSocket + A2A
shared authenticated entry points
MCP interface
read-oriented subset; not all host tools
Agent orchestrator
harness adapters · team channels
lead / workers · tasks · memory
host tools + scoped dispatch
shared node / team / run IDs
intent
Binding + approval bridge
node semantics · read/write scope
manual / auto policy · approvals
agent-write admission + readback
records · separate backstop
write
Industrial supervisor
DAQ runtimes + gateway sweep
DCW dispatch + protocol drivers
alarms · line / recipe / run state
device-twin state publication
External CLI engines
separate processes; containment external
Separate persistent stores
platform DB · time series · objects · audit
Plant / protocol boundary
physical devices or seeded scenarios